

TECHNICAL ASSESSMENT

Next Word Prediction System — Design Brief

For the role of Natural Language Processing Engineer

ISSUED BY	NexText AI — Language Systems Group
ROLE	NLP Engineer (Applied Machine Learning)
ASSESSMENT TYPE	Take-home design and implementation exercise
DATE OF ISSUE	30 June 2026
TIME ALLOWED	48 hours from receipt

1. CONTEXT

The candidate is being evaluated for a Natural Language Processing role at NexText AI, where engineering teams build production-grade language and sequence-modelling systems for a range of product and platform clients. This assessment presents a problem representative of the work the team undertakes, and is intended to evaluate the candidate's ability to design, implement, and clearly communicate a working solution.

2. PROBLEM STATEMENT

Design and implement a system that, given a sequence of words, learns the underlying patterns of natural language and predicts the most probable next word, producing coherent continuations of the supplied text.

3. OBJECTIVE

Given an input text sequence that exhibits one or more of the following characteristics:

- (a) a partial sentence or phrase whose continuation is to be predicted;
- (b) vocabulary, phrasing, or context not necessarily seen during training; or
- (c) variable sequence length and arbitrary starting position within a passage,

the system shall produce one or more next-word predictions that are grammatically plausible, contextually appropriate, and faithful to the language patterns learned from the training corpus.

4. FUNCTIONAL REQUIREMENTS

4.1 Input Handling and Tokenisation

The system shall accept raw text as input and perform the necessary pre-processing, including tokenisation, vocabulary construction, and conversion of tokens into integer sequences, without requiring additional user intervention.

4.2 Sequence Representation

The system shall transform tokenised text into fixed-length input sequences suitable for supervised next-word training, and shall represent words using a trainable embedding layer rather than sparse one-hot vectors.

4.3 Model Architecture

The system shall employ a recurrent neural architecture appropriate for sequential data. The candidate may use a simple RNN, a GRU, an LSTM, or a justified combination thereof, and shall be prepared to explain how the chosen architecture mitigates the vanishing gradient problem when modelling longer-range dependencies.

4.4 Prediction Behaviour

Given an input sequence, the system shall return the predicted next word, together with its associated probability where applicable. The system shall additionally support iterative generation, whereby successive predictions are fed back as input to generate a continuation of several words from a supplied seed text.

4.5 Generalisation

The system shall not be over-fitted to, or restricted to, the exact phrasing of the training corpus. It is expected to produce reasonable predictions for previously unseen but in-domain sequences, and to degrade gracefully when presented with out-of-vocabulary tokens.

5. INPUT AND OUTPUT SPECIFICATION

Input	A text sequence (a word, phrase, or partial sentence) supplied as a plain string, together with an optional parameter specifying the number of words to generate.
Output	The predicted next word for the given sequence and, where requested, the generated continuation text, returned as a standard string.

6. EVALUATION CRITERIA

Submissions will be assessed against the following criteria. Candidates are advised to address each criterion explicitly in their submission.

Criterion	Description
6.1 Prediction Accuracy	The quality and contextual appropriateness of the next-word predictions, evaluated qualitatively and, where appropriate, using metrics such as accuracy or perplexity.
6.2 Model Performance	Training and validation behaviour, including evidence of convergence and the absence of severe over-fitting or under-fitting.
6.3 Generalisation	The ability of the system to handle unseen sequences and out-of-vocabulary tokens.
6.4 Architecture Design	The appropriateness and correctness of the chosen RNN, GRU, or LSTM design, including the use of embedding layers.
6.5 Code Quality	The structure, readability, and reproducibility of the submitted implementation.
6.6 Clarity of Explanation	The candidate's understanding of the underlying concepts and the clarity with which the methodology is communicated.

7. SUBMISSION REQUIREMENTS

Candidates are required to submit, in addition to the working system, a short explanation video. The video is a mandatory component of this assessment and shall cover, at a minimum:

- (a) the candidate's overall approach to the problem and the rationale for the chosen technique;
- (b) a clear description of how the system works internally, including any pre-processing, tokenisation, embedding, model or algorithmic stages, and the prediction logic; and
- (c) a step-by-step dry run on at least one representative input sequence, demonstrating how the input is transformed at each stage of the pipeline.

Submissions that do not include the explanation video will be considered incomplete.

8. TIMELINE

The assessment shall be completed and submitted within **forty-eight (48) hours** from the time of issue. Late submissions will not be considered unless an extension has been agreed in writing in advance.

Note to candidate: You are encouraged to use any publicly available libraries, frameworks, or pre-trained models — including TensorFlow, Keras, NumPy, Pandas, Matplotlib, and NLTK or similar pre-processing libraries — provided that all such usage is disclosed in your explanation. You may select any reasonable text corpus for training; the choice of corpus and its preparation should be explained. Originality of approach is valued, but so is engineering judgement and clarity of communication. If any requirement in this document is unclear, please request clarification before commencing work.